

JavaScript Topics: Basic to Advanced

1. Basics & Fundamentals

- What is JavaScript?
- How JavaScript works (Execution Context, Call Stack)
- JS placement (inline, internal, external)
- Comments
- Variables (var, let, const)
- Data Types (Primitive & Non-Primitive)
- Type Conversion & Type Coercion
- Operators (Arithmetic, Comparison, Logical, Assignment, Ternary)

2. Control Flow

- if, else if, else
- switch
- Truthy & Falsy values
- Loops: for, while, do...while
- break and continue

3. Functions

- Function Declaration
- Function Expression
- Arrow Functions
- Parameters vs Arguments
- Default Parameters
- Return Statement

- Callback Functions
- Higher-Order Functions
- Function Hoisting

4. Scope & Execution

- Scope (Global, Local, Block)
- Lexical Scope
- Scope Chain
- Hoisting (Variables and Functions)
- Closures

5. Objects

- Object Literals
- Dot vs Bracket Notation
- Nested Objects
- Object Methods (this, shorthand syntax)
- Object.keys(), values(), entries()
- Object.assign(), Spread Operator
- Optional Chaining (?.)
- Destructuring (Object & Array)
- this keyword (arrow vs regular functions)

6. Arrays

- Creating Arrays
- Accessing elements
- Array Methods: push(), pop(), shift(), unshift(), splice(), slice(), forEach(), map(), filter(), reduce(), find(), some(), every(), sort(), reverse(), concat(), includes(), indexOf()

7. Advanced Functions

- Recursion
- First-Class Functions
- IIFE (Immediately Invoked Function Expressions)
- Memoization
- Debouncing & Throttling

8. Asynchronous JavaScript

- Synchronous vs Asynchronous
- setTimeout, setInterval
- Callback Hell
- Promises
- Promise Chaining
- Promise.all, Promise.race, Promise.allSettled
- Async/Await
- Error Handling with try...catch

9. DOM Manipulation (Browser Only)

- Selecting elements (getElementById, querySelector, etc.)
- Modifying content (innerText, innerHTML)
- Styling elements (style, classList)
- Event Listeners
- DOM Tree & Traversal
- Form handling & validation
- Event Bubbling & Capturing

10. Browser APIs

- localStorage, sessionStorage
- fetch() API
- XMLHttpRequest
- Geolocation API
- Clipboard API
- Notification API

11. Error Handling

- try, catch, finally
- throw custom errors
- Error object (name, message, stack)

12. ES6+ Modern JavaScript

- let, const
- Arrow Functions
- Template Literals
- Destructuring
- Spread & Rest Operators
- Modules (import/export)
- Enhanced Object Literals
- Optional Chaining (?.)
- Nullish Coalescing (??)
- Array.flat(), Array.from()

13. Advanced Concepts

- Closures

- this, call(), apply(), bind()
- Currying
- Event Loop
- Microtask vs Macrotask Queue
- Garbage Collection
- JavaScript Engines (V8, SpiderMonkey)
- Memory Management

14. OOP in JavaScript

- Constructor Functions
- Prototypes & Inheritance
- class & constructor
- extends & super
- Encapsulation, Inheritance, Polymorphism

15. Modules & Tooling

- ES Modules (import/export)
- CommonJS (Node.js)
- Bundlers (Webpack, Vite)
- Transpilers (Babel)
- Linting (ESLint)
- Prettier

16. Testing (Optional for Beginners)

- Manual Testing
- Console Logs
- Unit Testing with Jest

- Debugging with DevTools